Assignment - I
BS in Analytics and Sustainability Studies (2024-28)
Statistical Ecology
Tata Institute of Social Sciences, Mumbai

Assignment Weightage: 50%

Species Distribution Modelling using MaxEnt: Climate, Topography, and Vegetation Drivers in India

You will prepare spatial layers, run a MaxEnt model, evaluate its performance, and interpret the ecological meaning of the results

1. Dataset Provided:
Species occurrence data (CSV)
- One species per student (Pick any one from the dataset)

2. Data to be Downloaded by Students
a) Bioclimatic Variables
- Source: WorldClim (Version 2.1)
- Variables: BIO1–BIO19
- Resolution: ~1 km (30 arc-seconds)

b) Topographic Data (Source: SRTM / Copernicus DEM)

c) Vegetation Data
- Source: Sentinel-2 (Copernicus Open Access Hub / Google Earth Engine)
- NDVI (mean composite for one year or season)

Assignment Tasks:

1. Data Preparation & Spatial Processing
a) Occurrence Data Preparation
- Visualize species occurrence points on a map of India
- Check for duplicate records and spatial errors

b) Environmental Layers
- Clip all layers (bioclim, DEM and NDVI) to your species study extent
- Ensure:
  - Same projection
  - Same spatial resolution
  - Same extent
- Include:
  - A table listing all environmental variables used
  - Maps showing clipped layers

2. MaxEnt Modelling

a) Model Execution & Outputs

- Include:
  - Habitat suitability map
  - Response curves for key variables
  - Jackknife test of variable importance

- Explain:
  - Which variables contribute most?
  - What ecological processes do they represent?

3. Model Evaluation & Interpretation

Discuss:

- AUC value interpretation
- Strengths and weaknesses of AUC
- Any signs of overfitting
- What does the predicted distribution tell us?
- How could this model inform:
  - Conservation planning?
  - Climate change impact studies?
  - Invasive species risk?

Submission Guidelines

- Submit the report to TISSOL and all the outcomes from the MaxEnt Model (zipped file)
- Report Format: PDF and name your file as "*Name_Enrol No._<Species Name>*"
- Length: ~2500 - 3000 words, Font: Times New Roman, 12-point
- All figures, tables, and datasets must be properly cited
- Direct use of AI-generated text (including ChatGPT) will lead to supplementary assessment. The submission must reflect your own analysis and language.